Problem Statement - Username Validation System

Consider the following variable:

```
data = ("Level123", [1, 2, 3], {"1", "2", "3"})
```

Perform the following operations:

- Check whether the given string reads the same forward and backward (ignore case)
- If it is a palindrome → convert the entire string to uppercase, otherwise to lowercase, print.
- Delete the last element from the list and display it

Expected Output:-

```
False
level123
3
```

Problem Statement - Banking Transaction System

Consider the following variable:

```
data = ([1000, 2000, 3000], "AccX9", {500, 1000, 1500})
```

Perform the following operations:

- Update the second transaction by replacing it with half of its value
- If the string starts with an uppercase letter → convert it to lowercase, otherwise convert it to uppercase
- Remove the smallest value from the set
- Check whether the sum of the first two transactions is greater than the last transaction

---

Expected Output:-

```
[1000, 1000, 3000]
accx9
False
```

Problem Statement - Library Management System

Consider the following variable:

```
data = ([101, 102, 103, 104], "Book12", {101, 105})
```

Perform the following operations:

- Delete the first element from the list
- If the length of the string is greater than 5 → reverse it, otherwise convert it to uppercase
- Remove first value from the set(should not be printed)
- Check whether the length of the list is equal to the length of the string

---

Expected Output:-

```
[102, 103, 104]
21kooB
False
```

Problem Statement - Username Validation System

Consider the following variable:

```
data = ("Level123", [1, 2, 3], {"1", "2", "3"})
```

Perform the following operations:

- Check whether the string only (ignoring digits) reads the same forward and backward
- If it is a palindrome → convert the entire string to uppercase, otherwise to lowercase
- Count how many elements from the list (after converting to string) are present in the set

- Delete the last element from the list and display it

Expected Output:-

```
True
LEVEL123
3
3
```

Given: Data Processing System

```
data = ([1200, 2500, 3600, 4800], "UserZ12", {200, 400, 800, 1600})
```

Perform the following operations:

- Replace the third element of the list with the difference of first and second elements, print list
- Reverse the string only if its last character is a digit, otherwise keep it unchanged, print string
- Remove the largest value from the set
- Check whether the product of first two elements of list is divisible by the smallest element in set, and print the output.

Expected Output:

```
[1200, 2500, -1300, 4800]
21ZresU
True
```

Given: Secure Account Processing

```
data = ([1500, 3000, 4500, 6000], "AccSecure7", {250, 500, 1000, 2000})
```

Perform the following operations:

- Replace the second element of the list with the sum of first and last elements, print list
- If the string contains any digit → convert entire string to uppercase, otherwise lowercase, print string
- Remove the second largest value from the set
- Check whether the difference of last two elements of list is equal to any value in the set, print Boolean value

---

Expected Output:

```
[1500, 7500, 4500, 6000]
ACCSECURE7
False
```

Given: Transaction Analyzer

```
data = ([10, 20, 30, 40, 50], "TranX", {5, 10, 15, 20, 25})
```

Perform the following operations:

- Replace the middle element of the list with the product of first and last elements, print list
- If the string length is even → reverse it, otherwise keep it unchanged, print string
- Remove all elements greater than 15 from the set
- Check whether the updated middle element is divisible by the sum of remaining set values, print boolean value

---

Expected Output:

```
[10, 20, 500, 40, 50]
TranX
False
```

Question 1

Problem Statement

Consider the following variable:

```
data = ({"name": "Alice", "age": 25, "city": "NewYork"},
        "Python123Programming",
        [10, 20, 30, 20, 40, 20],
        {5, 10, 15, 20})
```

Perform the following operations:

- Extract all values from the dictionary and display them as a list
- Verify whether the string consists entirely of alphabetic characters and display the result
- Determine how many times the number `20` appears in the list
- Find the position of the first occurrence of `30` in the list
- Remove `10` from the set if it exists
- Extract characters from index 6 to 12 from the string

Expected Output:-

```
['Alice', 25, 'NewYork']
False
3
2
{5, 15, 20}
123Pro
```

Question 2

Problem Statement

Consider the following variable:

```
data = ({"product": "Laptop", "price": 1200, "brand": "Dell"},
        "   HelloWorld   ",
        [1, 2, 3, 4, 5],
        {100, 200, 300})
```

Perform the following operations:

- Display all key-value pairs from the dictionary
- Remove the extra spaces from the beginning and end of the string

- Add the elements `[6, 7, 8]` to the end of the list
- Remove and display the last element from the list
- Insert `400` into the set
- Check whether the string starts with `"Hello"`

Expected Output:-

```
[('product', 'Laptop'), ('price', 1200), ('brand', 'Dell')]
HelloWorld
[1, 2, 3, 4, 5, 6, 7, 8]
8
{100, 200, 300, 400}
True
```

Question 3

Problem Statement

Consider the following variable:

```
data = ({"math": 90, "science": 85, "english": 88},
        "DataScience2024",
        [15, 25, 35, 25, 45],
        {1, 2, 3, 4, 5})
```

Perform the following operations:

- Extract all keys from the dictionary
- Convert the string to lowercase and verify if it is in lowercase
- Count how many times `25` appears in the list
- Find the position of `45` in the list
- Remove `3` from the set
- Extract elements from index 1 to 4 with a step of 2 from the list

---

Expected Output:-

```
['math', 'science', 'english']
True
2
4
{1, 2, 4, 5}
[25, 25]
```

Question 4

Problem Statement

Consider the following variable:

```
data = ({"city": "Mumbai", "population": 20000000, "state": "Maharashtra"},
        "   PythonPro   ",
        [100, 200, 300, 400, 200],
        {10, 20, 30, 40})
```

Perform the following operations:

- Retrieve the value associated with the key `"city"`
- Remove all whitespace from the beginning and end of the string
- Add `500` to the end of the list
- Remove the first occurrence of `200` from the list
- Remove and display an arbitrary element from the set
- Extract characters from index 2 to the end, taking every second character from the string

---

Expected Output:-

```
Mumbai
PythonPro
[100, 200, 300, 400, 200, 500]
[100, 300, 400, 200, 500]
10  # or any random element from set
toPo
```

Question 5

Problem Statement

Consider the following variable:

```
data = ({"id": 101, "name": "John", "active": True},
        "HelloWorldPython",
        [5, 10, 15, 10, 20, 10],
        {50, 60, 70, 80})
```

Perform the following operations:

- Extract all values from the dictionary
- Check if the string is in uppercase
- Count how many times `10` appears in the list
- Find the position of `20` in the list
- Remove `60` from the set if it exists
- Reverse the entire string

Expected Output:-

```
[101, 'John', True]
False
3
4
{50, 70, 80}
nohtyPdlroWolleH
```

Question 6

Problem Statement

Consider the following variable:

```
data = ({"fruit": "apple", "color": "red", "price": 1.2},
        "12345",
        [2, 4, 6, 8, 10],
        {100, 200, 300, 400})
```

Perform the following operations:

- Display all key-value pairs from the dictionary
- Check if the string consists only of digits
- Remove all elements from the list
- Add `50` to the set
- Find the index of `8` in the list
- Extract the last 3 characters from the string

Expected Output:-

```
[('fruit', 'apple'), ('color', 'red'), ('price', 1.2)]
True
[]
{100, 200, 300, 400, 50}
3
345
```

Question 7

Problem Statement

Consider the following variable:

```
data = ({"a": 1, "b": 2, "c": 3, "d": 4},
        "PythonIsAwesome",
        [100, 200, 300, 400, 500],
        {1, 2, 3, 4, 5})
```

Perform the following operations:

- Extract and remove the last inserted key-value pair from the dictionary
- Convert the string to lowercase and check if it starts with `"python"`
- Remove the element `300` from the list
- Insert `600` at the end of the list
- Remove and display an arbitrary element from the set
- Extract every third character from the string starting from index 1

Expected Output:-

```
('d', 4)  # or last inserted item
{'a': 1, 'b': 2, 'c': 3}
True
[100, 200, 400, 500, 600]
1  # or any random element from set
yhnwsm
```

Question 8

Problem Statement

Consider the following variable:

```
data = ({"name": "Python", "version": 3.9, "year": 2024},
        "   SliceAndDice   ",
        [1, 1, 2, 3, 5, 8, 13],
        {10, 20, 30, 40, 50})
```

Perform the following operations:

- Retrieve the value for key `"version"` without causing an error if key doesn't exist
- Remove all leading and trailing spaces from the string
- Count how many times `1` appears in the list
- Insert `21` at index 3 in the list
- Remove `30` from the set
- Extract the first 5 characters from the string

Expected Output:-

```
3.9
SliceAndDice
2
[1, 1, 2, 21, 3, 5, 8, 13]
{10, 20, 40, 50}
Slice
```

Question 9

Problem Statement

Consider the following variable:

```
data = ({"country": "India", "capital": "Delhi", "language": "Hindi"},
        "Hello123World",
        [50, 40, 30, 20, 10],
        {7, 14, 21, 28})
```

Perform the following operations:

- Extract all keys from the dictionary
- Check if the string contains only alphabetic characters
- Sort the list in ascending order
- Remove the element `40` from the list
- Add `35` to the set
- Extract the substring from index 5 to 8 from the string

Expected Output:-

```
['country', 'capital', 'language']
False
[10, 20, 30, 40, 50]
[10, 20, 30, 50]
{35, 7, 14, 21, 28}
123
```

Question 10

Problem Statement

Consider the following variable:

```
data = ({"item": "Book", "quantity": 50, "price": 250},
        "ProgrammingFun",
        [25, 50, 75, 100, 125],
        {5, 10, 15, 20, 25})
```

Perform the following operations:

- Remove and return the last inserted key-value pair from the dictionary
- Convert the string to uppercase and check if it ends with `"FUN"`
- Extend the list with `[150, 175]`
- Find the index of `75` in the list
- Remove `15` from the set using a method that won't raise an error if it doesn't exist
- Extract elements from index 2 to the end of the list with a step of 2

Expected Output:-

```
('price', 250)  # or last inserted item
{'item': 'Book', 'quantity': 50}
True
[25, 50, 75, 100, 125, 150, 175]
2
[75, 125, 175]
```